

Semester Thesis

Understanding system boundary restrictiveness
and solution robustness in energy system models

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Abstract

Read the paper "A guide to writing articles in energy science" from Martin Weiss & Alexandra M. Newman (2011)

Abstract and Introduction The abstract (short summary) is placed at the very beginning of the work. It comprises at most one page. In the abstract, only the following questions are of interest: What is your work about (question)? Which methods were used (solution)? What is the outcome (results)? A reader who has read the abstract should know the rough outlines of the work and its most important results (no details!). Not included in the abstract are the motivation, history of the problem, or information about the structure of the work Usually, the abstract does not contain equations, figures, tables, literature citations, or other references to other pages of the paper The introduction marks the beginning of the actual work. It leads to the research question by showing the bigger picture of the research and placing it in relation to existing knowledge (literature). In contrast to the abstract, the introduction, therefore, also includes statements on the motivation and background of the work. The introduction then defines the scientific questions to be examined in the context of the work and mentions the methods used or further developed to solve the problem. The last section of the introduction gives an overview of the structure of the work ("storyline"). A brief description suffices of the steps of the research described in each chapter and how the chapters build on each other. However, no results are presented yet. Only the procedure is explained!

Acknowledgments

Not mandatory.

Contents

Abstract	i
Acknowledgments	ii
List of Tables	v
List of Figures	v
List of Abbreviations	v
List of Mathematical Symbols	v
1 Introduction and Motivation	1
1.1 Motivation	1
1.2 Contributions	1
2 Background	2
2.1 Linear Programming Duality and Sensitivity	2
2.2 Solving Linear Programs	2
2.3 Boundary Information in Established Energy System Models	3
3 Framework: ZEN-garden and the Crystal Ball Dataset	4
3.1 ZEN-garden	4
3.2 The Crystal Ball Dataset	4
4 Methodology and Implementation	5
4.1 Capex-Equivalent Reduced Cost from the Lifetime Dual	5
4.2 Shadow-Price Extraction and the Generic-Technology Trick	5
5 Results	6
5.1 Test Models	6
5.2 Case Study: Reduced Costs	6
5.3 Case Study: Shadow Prices	6
6 Discussion	7

7	Conclusion and Outlook	8
	Appendix	9
A1	Numerical Pitfalls of Value Extraction	9
A1.1	Degeneracy: Unreliable Duals and Reduced Costs	9
A1.2	Presolve Breaks Dual Recovery	9
A1.3	True Dual Degeneracy	9
A2	Solver Configuration	9
A3	Derivation of the Capex-Equivalent Reduced Cost	10

List of Tables

List of Figures

List of Abbreviations

Acronyms

CAPEX capital expenditure

CRF capital recovery factor

ESM energy system model

RC reduced cost

SP shadow price

List of Mathematical Symbols

C_{Fuel} Fuel costs

1 Introduction and Motivation

1.1 Motivation

- Cost-optimal energy system models (ESMs) return a single least-cost solution.
- Decision makers care about more than the optimum: robustness, which technologies are “almost” competitive, and what a constraint costs at the margin.
- LP duality already encodes this in reduced costs and shadow prices, but ESMs typically discard it.

1.2 Contributions

- Review of how boundary information is handled in established ESMs.
- A post-processing pipeline that extracts shadow prices and reduced costs from ZEN-garden.
- A capex-equivalent reduced-cost metric for investment variables.
- A characterisation of the numerical pitfalls (degeneracy, presolve, scaling) that make naive dual readings unreliable, with mitigations.

2 Background

2.1 Linear Programming Duality and Sensitivity

- State the primal LP $\min c^\top x$ s.t. $Ax \geq b$, $x \geq 0$ and its dual.
- Introduce the basis: with m equality constraints, a basic solution has exactly m basic variables and the basis matrix is invertible.

Basic vs. non-basic variables and complementary slackness.

- A basic variable has RC zero by definition; a variable with non-zero RC is provably non-basic.
- Complementary slackness links primal activity and dual constraints.

Shadow price.

- The SP of a constraint is the marginal change of the objective per unit relaxation of that constraint's right-hand side.
- Valid only within the basis-stable interval.

Reduced cost.

- The RC of a (currently non-basic) variable is the amount by which its objective coefficient must improve before it enters the optimal basis at a positive level.
- Central reading: “how much cheaper must this technology become to be built”.

2.2 Solving Linear Programs

Simplex (primal and dual).

- Vertex-following methods.
- Always terminate at a basic (vertex) solution, hence return a basis and well-defined RC/SP.

Barrier / interior point.

- Follow the central path to an interior optimum.
- On their own they return a non-vertex solution, so basis-dependent duals are not directly available.

Crossover.

- Pushes a barrier solution to an adjacent vertex so a basis – and therefore exact RC/SP – becomes available.
- Required for the VBasis flag used in Section 4.

Presolve and scaling.

- Presolve aggregates/substitutes constraints and variables for speed, which can make certain constraint duals unrecoverable in the original space.
- Scaling rescales rows/columns and must be undone to report duals in input units.
- Both effects motivate the robustness study in Section 5.

2.3 Boundary Information in Established Energy System Models

tbd

3 Framework: ZEN-garden and the Crystal Ball Dataset

3.1 ZEN-garden

- Open-source framework for optimising energy transition pathways with user-oriented data handling.
- The optimisation model is assembled with `linopy` and handed to Gurobi.

Investment variables and the capacity coupling.

- `capacity_addition` is the investment decision.
- `constraint_technology_lifetime` is an equality coupling installed `capacity` to the sum of additions over the lifetime window.
- Because it is an equality, at least one coupled variable must be basic per (technology, node, year) – the structural source of the degeneracy discussed in Section 5 and Appendix A1.

3.2 The Crystal Ball Dataset

- 28-node European electricity-system instance.
- Temporal scope: planning horizon to 2050.
- Give the technologies considered and the node set used for the per-country results.

4 Methodology and Implementation

4.1 Capex-Equivalent Reduced Cost from the Lifetime Dual

- The native solver RC of `capacity_addition` is in annualised objective units and is often zero for degenerate-basic variables.
- Instead, reconstruct an investment-level signal from the dual of the lifetime coupling constraint.

The capex-equivalent reduced cost is

$$\text{rc_capex_equivalent} = \text{capex_specific} - \frac{\sum (-\text{dual}_{\text{lifetime}})}{\text{annuity} \cdot \sum \text{discount_factor}}, \quad (1)$$

reported in input units (€/kW).

- Derive the annuity / capital recovery factor (CRF) factor.
- Show the bit-exact identity for a marginally-built technology (lifetime dual = $-(\text{capex} \cdot \text{CRF})$); full derivation in Appendix A3.

4.2 Shadow-Price Extraction and the Generic-Technology Trick

- Shadow prices are read directly from constraint duals.
- To isolate the marginal value of a specific boundary, insert a *generic technology* whose single defining constraint exposes the desired SP in isolation.

5 Results

5.1 Test Models

We evaluate the implementation on two models.

- **Toy LP.** A small, hand-checkable model where RC/SP are known in closed form. Used to verify correctness, including the perturbation check (reducing a technology’s capital expenditure (CAPEX) by exactly its capex-equivalent RC brings it to the build margin).
- **Crystal Ball.** The 28-node European instance (Section 3.2). Used to apply the method at scale, with cloned/perturbed technologies to test the “CAPEX – RC → just gets built” behaviour.

5.2 Case Study: Reduced Costs

tbd

5.3 Case Study: Shadow Prices

tbd

6 Discussion

tbd

7 Conclusion and Outlook

- Contributions
- Outlook

Appendix

A1 Numerical Pitfalls of Value Extraction

A1.1 Degeneracy: Unreliable Duals and Reduced Costs

- Degeneracy in the LP affects *both* families of sensitivity information: constraint duals (shadow prices) and variable reduced costs.
- The lifetime equality forces a coupled variable into the basis; the crossover tie-break often makes `capacity_addition` basic at value zero, so its native RC reads as zero by definition even when the technology is not built.
- The same non-uniqueness can make the recovered dual basis-dependent, so a naive read of either quantity can be misleading.

A1.2 Presolve Breaks Dual Recovery

- With presolve on, the dual of `constraint_technology_lifetime` is not recoverable in the original variable space.
- Fix: set `Presolve` 0.

A1.3 True Dual Degeneracy

- Non-unique lifetime dual makes the capex-equivalent RC basis-dependent. Case: Value is zero, but also the manually calculated capex-equivalent RC is zero, so the technology looks like being on the verge of being built.
- A small CAPEX perturbation flips vertices and yields a non-monotonic response (still not built).

A2 Solver Configuration

- Recommended Gurobi settings [...] for reliable dual and RC recovery.

A3 Derivation of the Capex-Equivalent Reduced Cost

- Further derivation of formulas



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